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Networking.

By Yuliana Martinez

Agenda

01 Key Concepts of Connectivity and Communications

02 Network Types

03 Elements of Networks

04 Network Reference Models

05 Switching Concepts

06 Routing Concepts

07 Common Networking Protocols

08 IPSEC VPN / SSL VPN

Chapter

01

Key Concepts of Network Connectivity and Communications

Device.

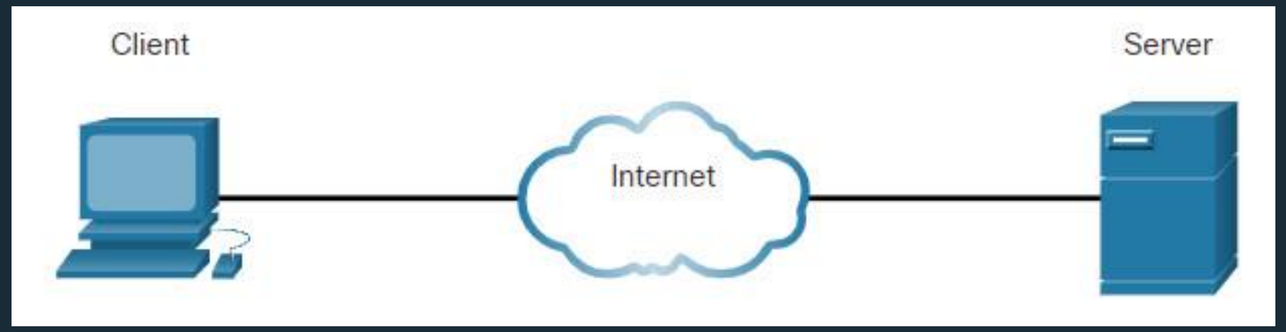
The term *device* is used to mean any piece of electronic equipment that has a network interface card of some sort.

End-user devices or consumer devices such as personal computer, laptops, smartphones, tablets, and printers as well as infrastructure devices such as switches, routers, firewalls, and load balancers have been interconnected in both private and public networks for many years.

Host.

All computers that are connected to a network and participate directly in network communication are classified as hosts. Hosts can be called end devices. Some hosts are also called *clients*. However, the term hosts specifically refers to devices on the network that are assigned a number for communication purposes. This number identifies the host within a particular network. This number is called the Internet Protocol (IP) address. An IP address identifies the host and the network to which the host is attached.

Servers are computers with software that allow them to provide information, like email or web pages, to other end devices on the network. Each service requires separate server software. For example, a server requires web server software in order to provide web services to the network. A computer with server software can provide services simultaneously to many different clients.



Network Architecture.

Network Security and Reliability

•**Security:** Protects data confidentiality, integrity, and access. Unauthorized access or ex posure of sensitive data violates security principles.

•**Fault Tolerance:** Achieved by providing redundant links and devices to maintain network availability.

•**Scalability:** Ensures the network can expand without degrading service for existing user s.

•**Quality of Service (QoS):** Prioritizes critical traffic like video conferencing to maintain p erformance.

•**BYOD (Bring Your Own Device):** Trend allowing personal devices to access corporate n etworks, requiring careful security management.

Types of Computer Network Architecture

Computer Network falls under these broad Categories:

•**Client-Server Architecture:** Client-Server Architecture is a type of Computer Network Architecture in which Nodes can be Servers or Clients. Here, the server node can manage the Client Node Behaviour.

•**Peer-to-Peer Architecture:** In P2P (Peer-to-Peer) Architecture, there is not any concept of a Central Server. Each device is free for working as either client or server.

UDP Operations.

which stands for “User Datagram Protocol”

The most common LAN technology is Ethernet. Ethernet is a data link layer protocol defined by the IEEE.

Latency is another concern with UDP traffic. Most applications that use UDP send small packets of interactive traffic such as voice.

The Ethernet protocol has evolved from transfer rates of 10 Mbps to 100 Gbps and even 400 Gbps.

Latency is another concern with UDP traffic. Most applications that use UDP send small packets of interactive traffic such as voice.

TCP Operations.

which stands for “Transmission Control Protocol”

TCP, as a connection-oriented protocol, uses fields, in its header to negotiate various settings with its peers.

The PDU at the data link layer is called a *frame*.

The Ethernet protocol has evolved from transfer rates of 10 Mbps to 100 Gbps and even 400 Gbps.

The data traffic between devices connected to the LAN can be transferred by a *switch*.

Chapter

02 Network Types

What Are Networks?

NIST CNSSI 4009 defines a network as follows:

"Information system(s) implemented with a collection of interconnected components.

Such components may include routers, hubs, cabling, telecommunications controllers, key distribution centers, and technical control devices."

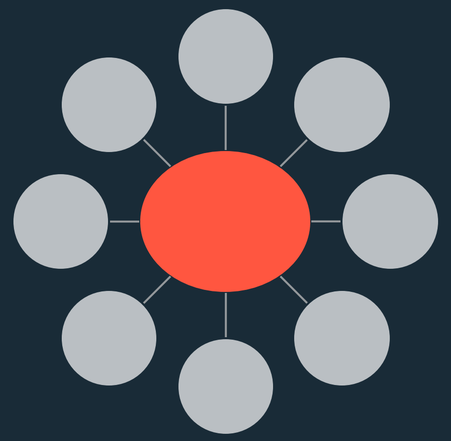
Three different factors are used to classify networks:

- Topology
- Standard
- *Size*

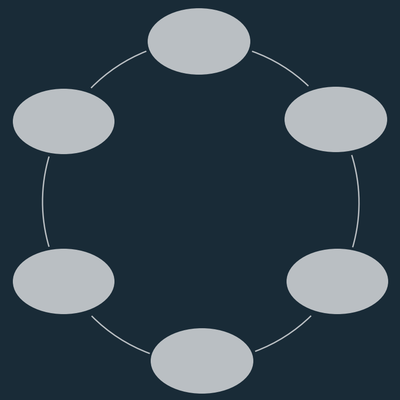
Topology.

Refers to how the various nodes or elements connect.

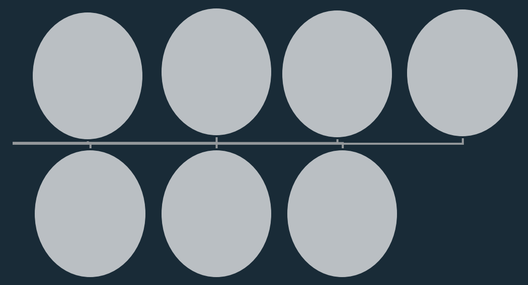
Star: All nodes in the system are connected to a central point.



Ring: is very similar to a star except that a token is passed around, and a node can transmit only when it has a valid token.



Bus: all the elements are connected one after the other.



Standard.

Refers to the protocol that the network uses.

- Ethernet is one example of a standard. An Ethernet network uses a star topology with concentrators connecting all the nodes in the network.
- Another standard is Token Ring, which, as the name suggests, is a ring topology type.

Size.

Local-area network (LAN):

A LAN is typically a single network for an office, an enterprise, or a home, and it is located completely within a single facility.

Campus-area network (CAN)

A CAN consists of two or more LANs within a limited area. CANs help interconnect various buildings or departments within a campus of an enterprise or large office or school.

Metropolitan-area network (MAN)

A MAN is a network that is extended to cover multiple locations within a slightly larger distance than CAN. A classic example of a MAN is the cable network in a city.

Wide-area network (WAN)

A WAN covers an even larger geographic area than a MAN. The connections are through public networks, such as the telephone system, microwaves, satellite links or leased lines.

Chapter

03

Elements of Networks

A network has several basic components:

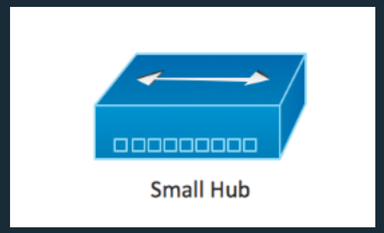
- Hubs
- Switches
- Bridges
- Routers

Hubs

A hub is the least intelligent and most rudimentary network device.

With a hub, traffic or data packets that come in any port are always sent out to all other ports.

Every Device Receives Every Packet.

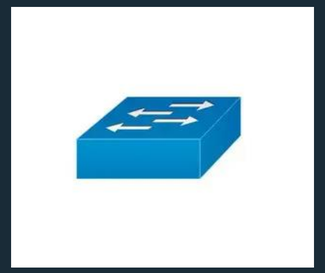


Switches

You can think of a switch as an “intelligent” hub.

A switch does what a hub does – but more efficiently and intelligently. It analyses the traffic as it flows.

A switch builds a port-to-MAC address table as devices start communicating.



Bridges

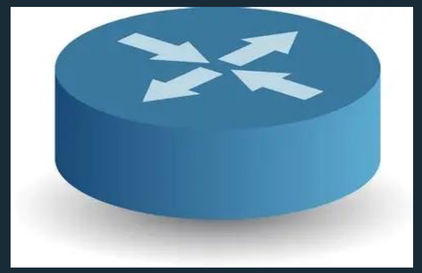
A bridge connects two physical network segments that use the same protocol.

Each network bridge builds a MAC address table of all devices that are connected on its ports.

Routers

A router is a device that forwards packets between networks via the network layer of the OSI model (Layer 3).

A router also has the intelligence to determine the best path to reach a particular network or device.



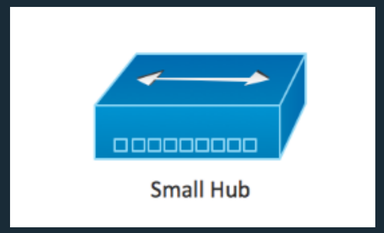
Gateway

A Gateway connects two different networks that use different protocols.

Translates data between different systems

Enables communication between dissimilar networks

Commonly used to connect private networks to external networks



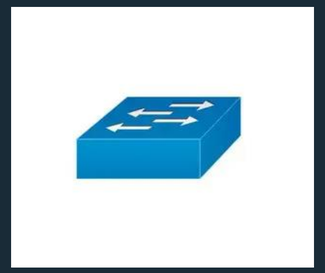
Access Point (AP)

An Access Point provides wireless connectivity to devices in a network.

Extends a wired network into Wi-Fi

Allows mobile devices to connect wirelessly

Improves network coverage area



Modem

A Modem converts digital data into signals suitable for transmission and vice versa.

Connects a home or office network to the ISP

Converts digital signals to analog and back

Enables internet access

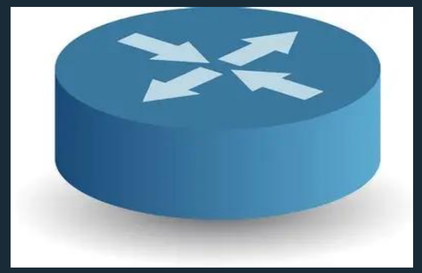
Firewall

A Firewall is a security device that monitors and controls network traffic.

Blocks unauthorized access

Filters incoming and outgoing data

Protects networks from cyber threats



Chapter

04

Network Reference Models

The OSI Model.

In the layered approach, each layer provides a set of functionality and features to the layer above it and consumes the features of the layer below it.

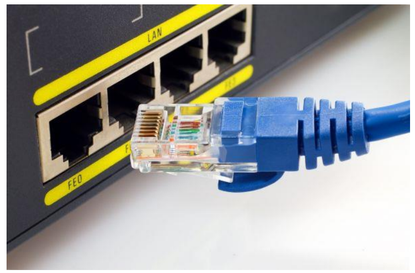
The OSI model defines seven layers for exchanging data between networked devices.

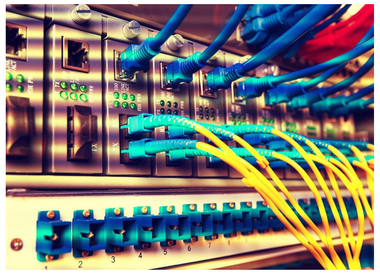
Two or more devices that would exchange data with each other using the OSI Model need to implement all seven layers.

Layer 1.

The physical layer is the first and lowest layer in the OSI model of computer networking. It is responsible for transmitting raw data bits through physical network hardware, converting data into signals that can travel through wires, fiber optics, or wireless channels, and turning these signals back into data at the receiver.

- Handles the **transmission and reception of raw bit streams** over physical media.
- Defines electrical, optical, and mechanical interfaces, including cables, connectors, and signaling methods.
- Devices: Hubs, repeaters.
- Example: Ethernet cables, fiber optics.





Application

Presentation

Session

Transport

Network Layer

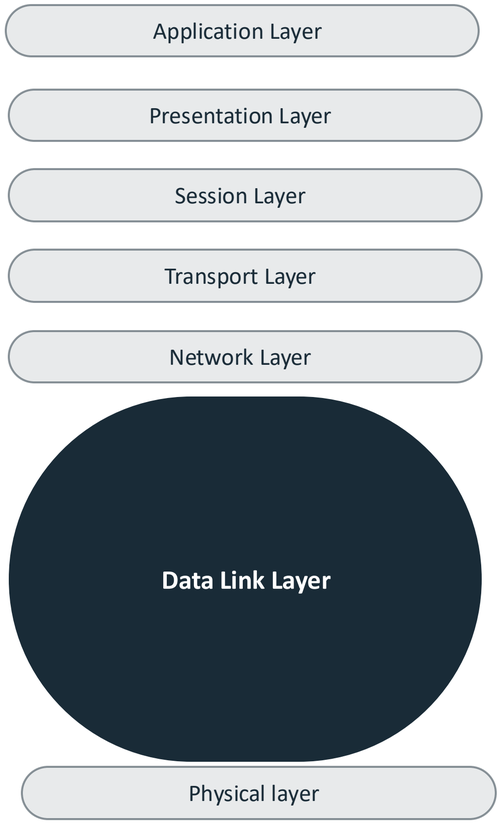
Data Link

Physical Layer

Layer 2.

The data link layer provides error-free transfer of data frames from one node to another over the physical layer, allowing layers above it to assume virtually error-free transmission over the link.

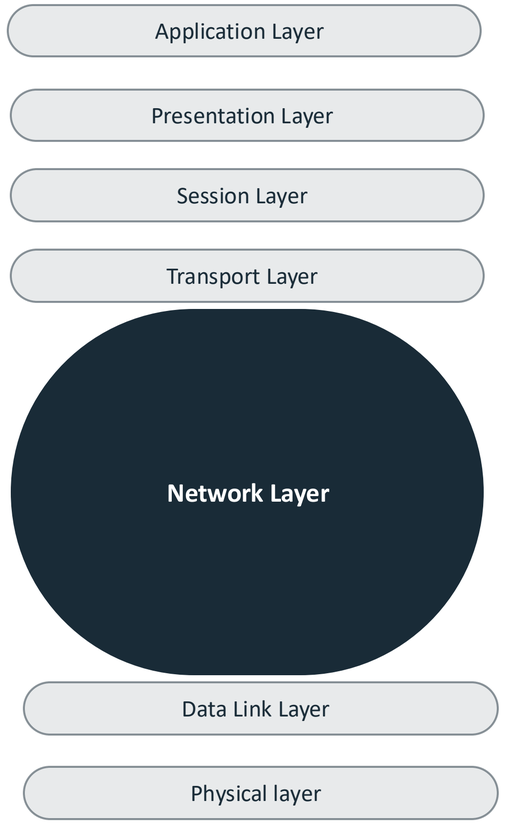
- Ensures error-free transfer of data frames between nodes on the same network.
- Uses MAC addresses for node-to-node communication.
- Devices: Switches, bridges.
- Protocols: Ethernet, PPP, VLAN tagging.



Layer 3.

The network layer controls the operation of the subnet, deciding which physical path the data should take based on network conditions, priority of service, and other factors.

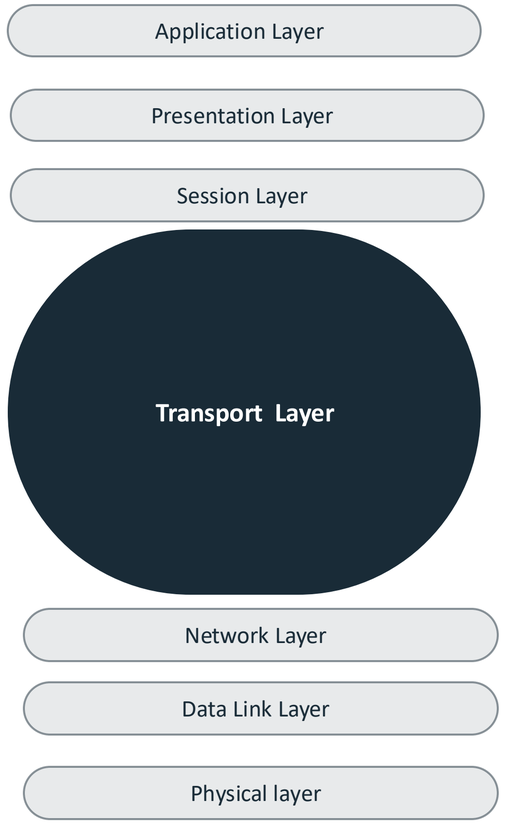
- Manages logical addressing and routing of data across networks.
- Determines the best path for data packets based on network conditions.
- Devices: Routers.
- Protocols: IP (IPv4, IPv6), ICMP.



Layer 4.

The transport layer ensures that messages are delivered error-free, in sequence, and with no losses or duplications.

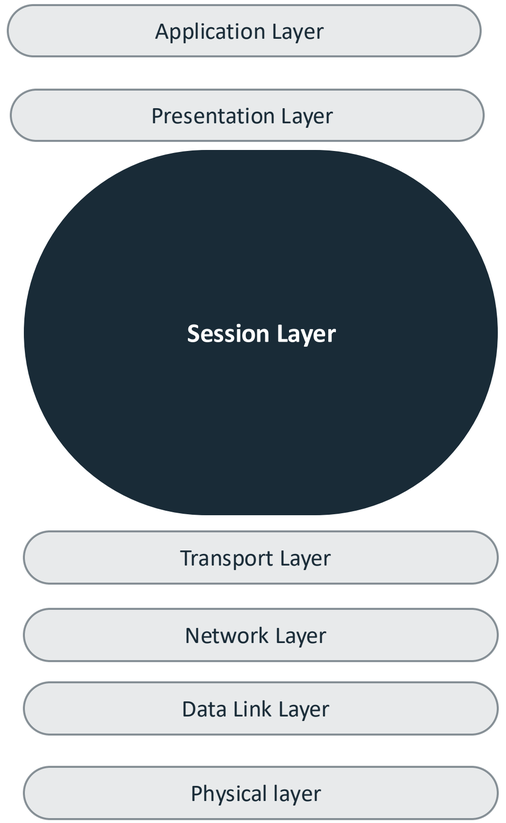
- Provides end-to-end communication, reliability, and flow control.
- Segments messages, ensures correct sequencing, and handles retransmissions.
- Protocols: TCP, UDP.



Layer 5.

The session layer allows session establishment between processes running on different stations.

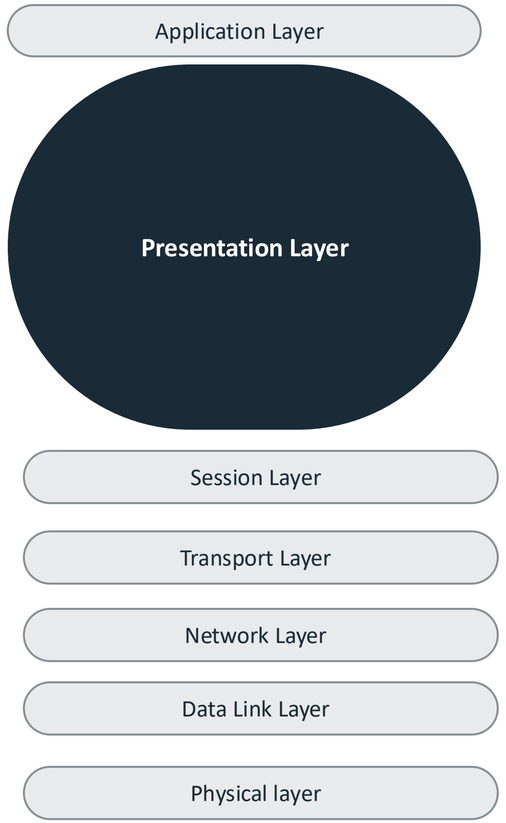
- Manages sessions between applications, including establishing, maintaining, and terminating connections.
- Protocols: NetBIOS, RPC, PPTP.



Layer 6.

The presentation layer formats the data to be presented to the application layer. It can be viewed as the translator for the network. This layer may translate data from a format used by the application layer into a common format at the sending station, then translate the common format to a format known by the application layer at the receiving station.

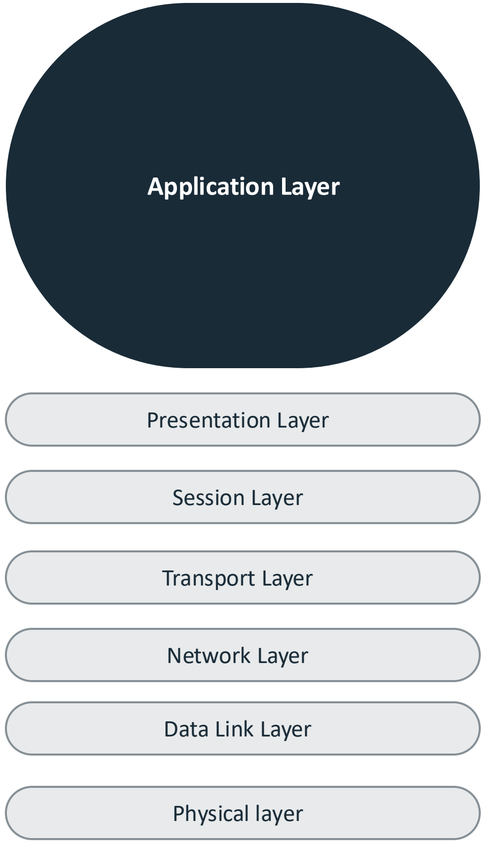
- Acts as a translator, formatting data for the application layer.
- Handles encryption, compression, and data conversion between different formats.
- Example: ASCII to EBCDIC conversion, SSL/TLS encryption.



Layer 7.

The application layer serves as the window for users and application processes to access network services.

- Provides network services directly to end-user applications.
- Interfaces with software like web browsers, email clients, and file transfer programs.
- Protocols: HTTP, FTP, SMTP, DNS.



The TCP/IP Model.

The TCP/IP model is a set of protocols that allows different types of computers to communicate over a network. It is the foundation of the Internet and is widely used in various networking environments. The model is divided into four layers, each with specific functions and protocols.

The TCP/IP Model.

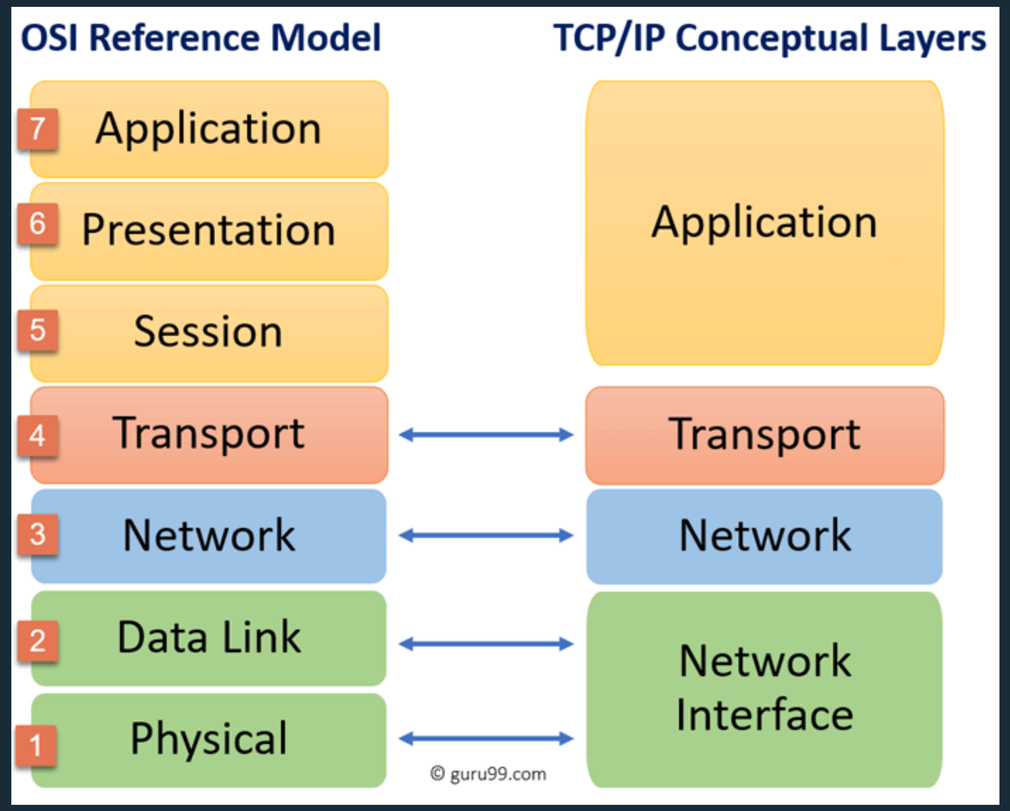
Layers of the TCP/IP Model

1. Application Layer: This layer provides network services directly to the user's applications. It includes protocols such as HTTP, FTP, SMTP, and DNS. These protocols enable web browsing, file transfer, email communication, and domain name resolution.

2. Transport Layer: This layer is responsible for end-to-end communication and data transfer between devices. The main protocols in this layer are TCP (Transmission Control Protocol) and UDP (User Datagram Protocol). TCP is a connection-oriented protocol that ensures reliable data transfer through error checking and flow control. UDP is a connectionless protocol that provides faster but less reliable data transfer.

3. Internet Layer: This layer handles the routing of data packets across the network. The primary protocol in this layer is IP (Internet Protocol), which is responsible for addressing and routing packets to their destination. Other protocols in this layer include ICMP (Internet Control Message Protocol) for error reporting and diagnostics, and ARP (Address Resolution Protocol) for mapping IP addresses to MAC addresses.

4. Network Access Layer: This layer is responsible for the physical transmission of data over the network. It includes protocols and technologies such as Ethernet, Wi-Fi, and PPP (Point-to-Point Protocol). This layer ensures that data is transmitted over the physical medium, such as cables or wireless signals.



Chapter

05

Switching Concepts

Data Frames switching is a critical process in moving data traffic from a source network endpoint to a destination endpoint within a local-area network (LAN).

Ethernet.

The most common LAN technology is Ethernet. Ethernet is a data link layer protocol defined by the IEEE.

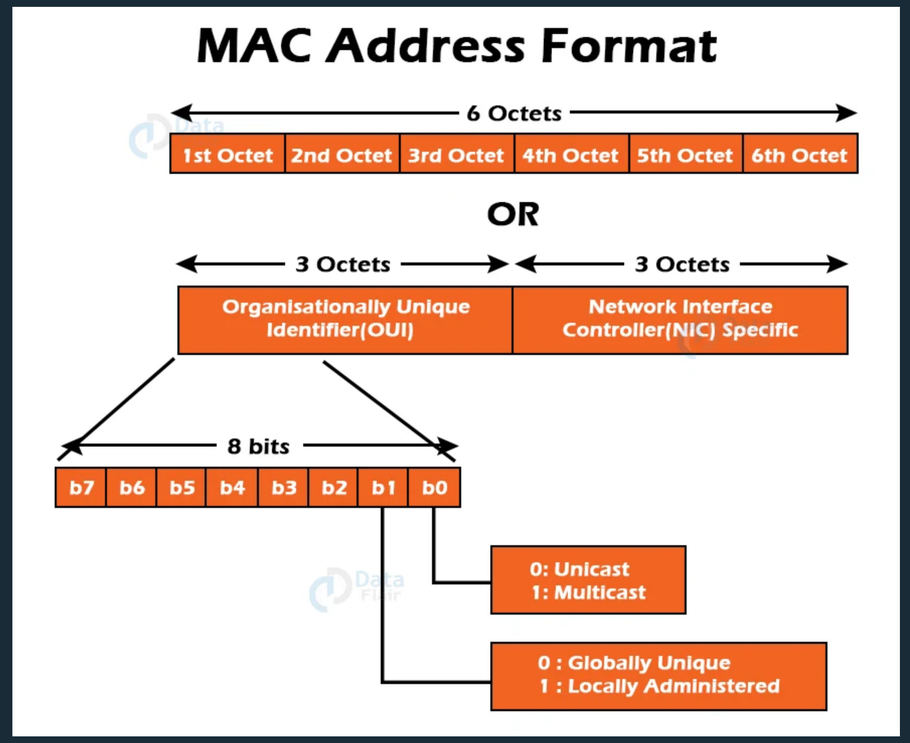
The PDU at the data link layer is called a *frame*.

The Ethernet protocol has evolved from transfer rates of 10 Mbps to 100 Gbps and even 400 Gbps.

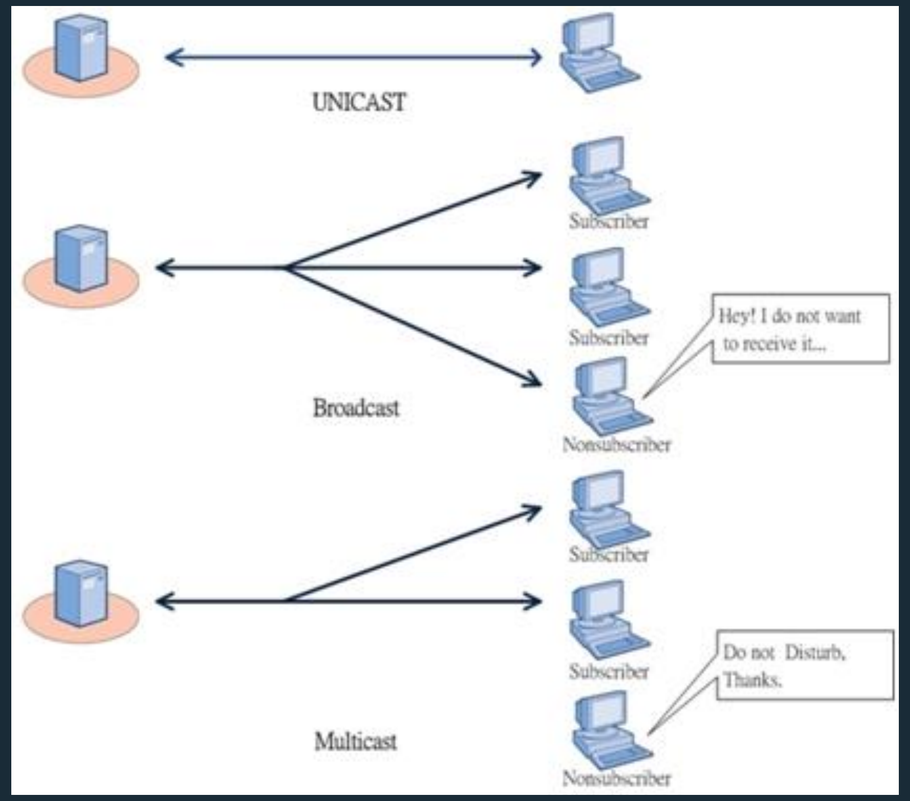
The data traffic between devices connected to the LAN can be transferred by a *switch*.

MAC Addresses.

Media Access Control (MAC) addresses are used to enable communications between devices connected to a local network.



MAC Addresses.



Unicast: This method is used for sending data to a single destination. It is the most common type of traffic in TCP/IP networks, such as when accessing a web server.

Broadcast: This method sends data to all devices on a network segment. It is used to announce something to all devices on the segment, such as ARP (address resolution protocol).

Multicast: Unlike unicast, which sends data to a single destination, multicast sends data to multiple interested receivers. This method is used for sending data to a group of hosts, such as in video over IP communication.

Virtual Local-Area Networks (VLANs).

A VLAN is a group of devices on an Ethernet network that are logically segmented by function, team, or application.

Each VLAN defines its own broadcast domain.

No matter the physical location in the network, devices can be part of the same VLAN.

Each VLAN defines its own broadcast domain.

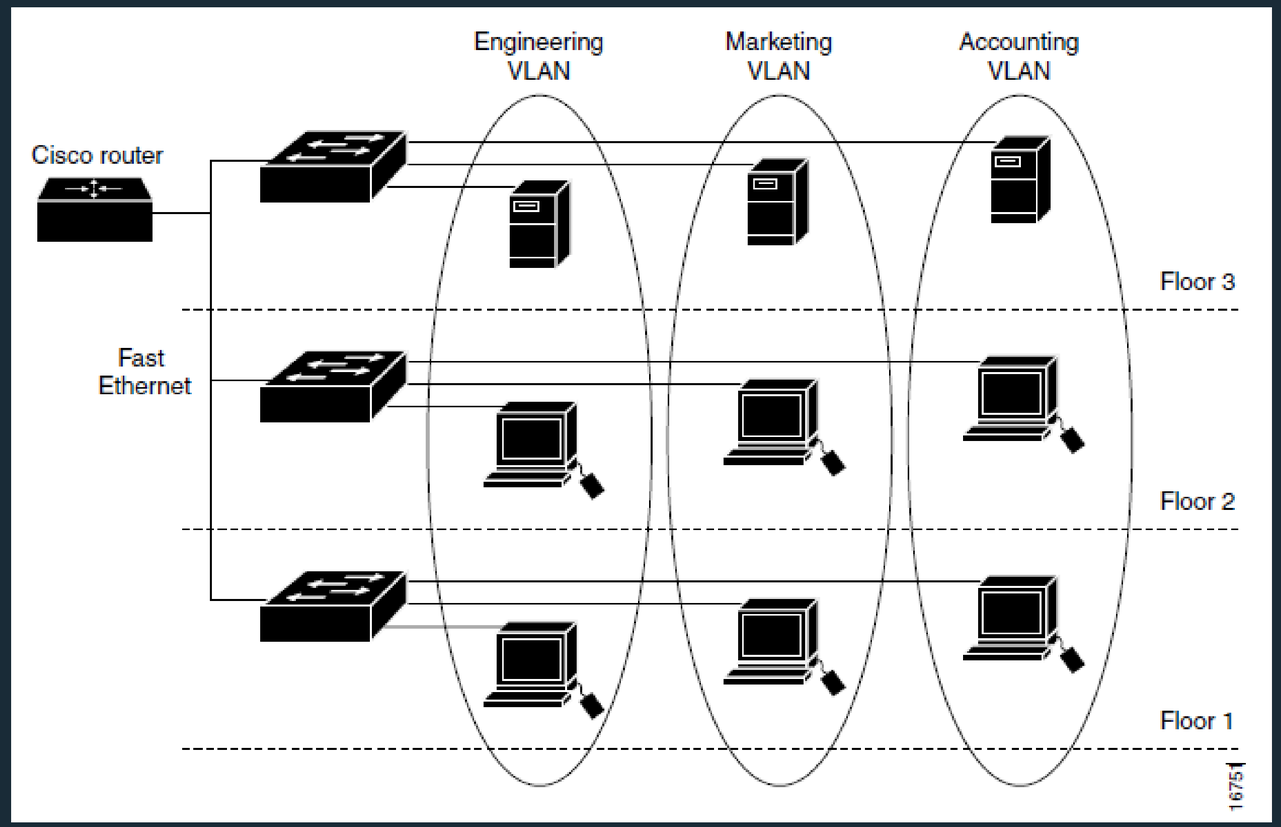
VLAN identifier, a maximum of 4094 VLANs can be defined; IDs 0 and 4095 are reserved.

Virtual Local-Area Networks (VLANs).

VLAN 10 – Engineering VLAN

VLAN 20 – Marketing VLAN

VLAN 30 – Accounting VLAN



Switching.

Switching is the process through which a data frame is forwarded from its source toward its destination by a Layer 2 device called a switch.

In a typical LAN, all devices connect to the network either through an Ethernet port available on the NIC or through wireless NIC.

The switch dynamically builds and maintains the MAC address table and indicates where each device is connected in the network.

A MAC address table on a switch might look as shown in the table:

MAC Address Table

VLAN	MAC Address	Type	Ports
10	001b.10a0.2500	Dynamic	Gi0/1
20	001b.10ae.7d00	Dynamic	Gi0/2
30	0050.7966.6705	Dynamic	Gi0/3

Chapter

06

Routing Concepts

Routing, or Layer 3 packet forwarding, is the process of selecting a path through a network.

IPv4 Addresses.

In networking, IPv4 addresses are structured as 32 bits long, divided into two components: the network portion and the host portion. The Network ID (NetID) is a fixed bit in the IP address that represents the entire network of any host connected to the network. The Host ID (HostID) is the part of the IP address that uniquely identifies a specific device within that network. The IP address is typically represented in dotted-decimal notation, where 32 bits are divided into four octets. Each octet can be represented in decimal format, separated by decimal points.

IPv4 Address Representation with Decimal and Binary Values.

Decimal	192	168	48	65
Octet/binary	11000000	10101000	00110000	01000001

Private IPv4 Addresses.

Address Class	Private Network ID	Network Address Range
A	10.0.0.0	10.0.0.0 to 10.255.255.255
B	172.16.0.0	172.16.0.0 to 172.31.255.255
C	192.168.0.0	192.168.0.0 to 192.168.255.255

IPv6 Addresses.

Internet Protocol version 6 (IPv6) expands the number of network address bits from 32 bits (in IPv4) to 128 bits, which provides more than enough globally unique IP addresses for every networked device on the planet. The unlimited address space provided by IPv6 allows Cisco to deliver more and newer applications and services with reliability, improved user experience, and increased security.

IPv6 Address Type	Preferred Format	Compressed Format
Unicast	2001:0:0:0:DB8:800:200C:417A	2001::DB8:800:200C:417A
Multicast	FF01:0:0:0:0:0:0:101	FF01::101
Loopback	0:0:0:0:0:0:0:1	::1
Unspecified	0:0:0:0:0:0:0:0	::

Subnetting.

Subnetting divides a network into smaller subnets, CIDR allows flexible IP address allocation without class restrictions, and VLSM enables subnets of varying sizes to optimize IP usage.

CIDR (Classless Inter-Domain Routing)

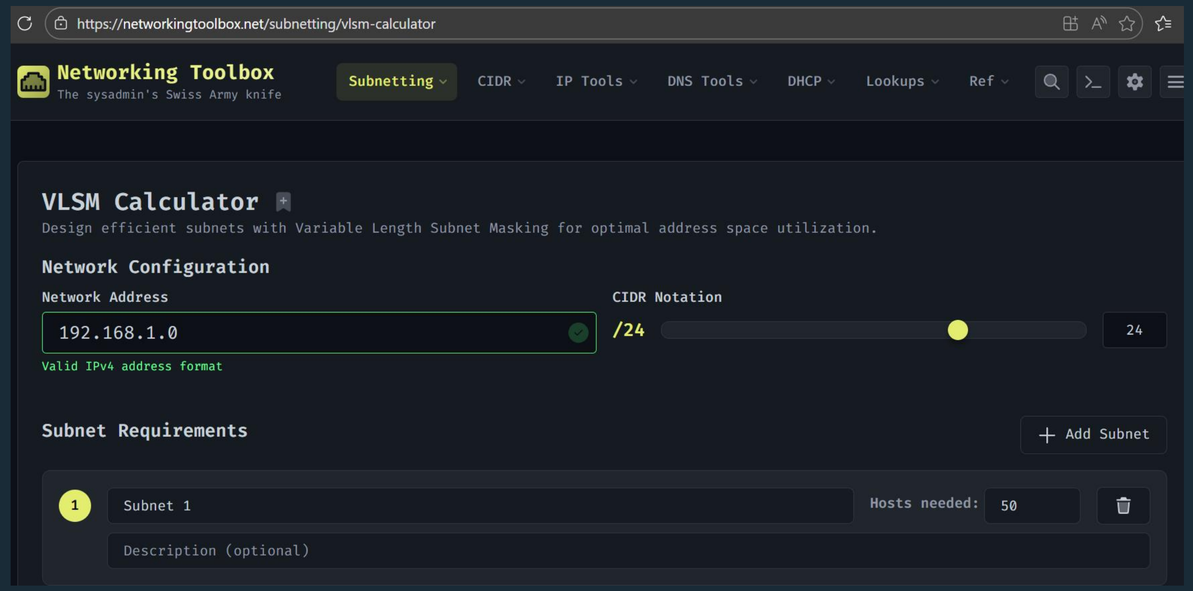
CIDR is a method of IP addressing that eliminates the traditional class-based system (Class A, B, C) and allows networks to be defined with variable-length prefixes, such as /24, /25, or /30. CIDR improves routing efficiency by enabling IP address aggregation, reducing the size of routing tables, and allowing more precise allocation of IP addresses for networks of different sizes. CIDR notation specifies the number of bits used for the network portion of the address, e.g., 192.168.1.0/24 means the first 24 bits are the network part.

VLSM (Variable Length Subnet Mask)

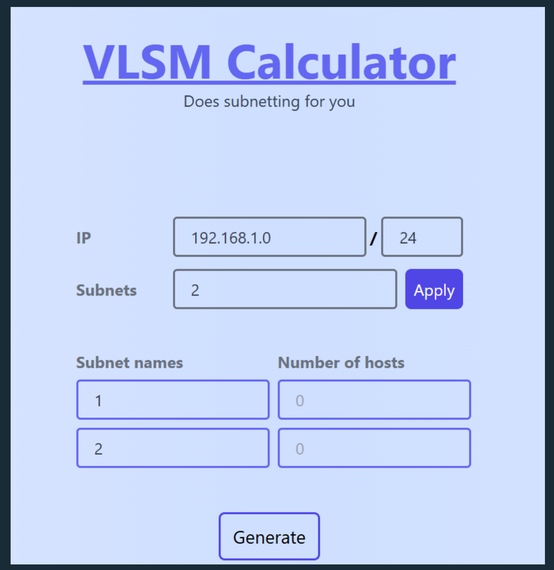
VLSM is an advanced subnetting technique that allows subnets within the same network to have different subnet masks based on actual host requirements.

Unlike fixed-length subnetting (FLSM), which wastes IP addresses by forcing all subnets to be the same size, VLSM reduces IP waste by creating subnets that match the number of hosts needed. For instance, a /24 network can have one /25 subnet for 120 hosts, a /28 subnet for 14 hosts, and a /30 subnet for a point-to-point link, all within the same network.

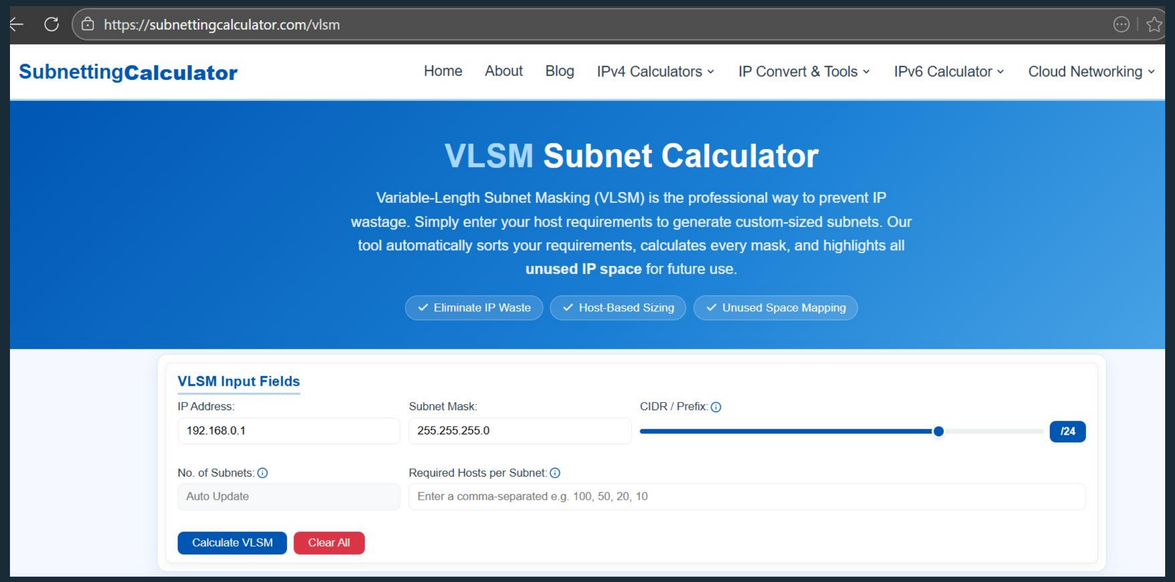
VLSM Calculator | Networking Toolbox



VLSM Calculator



VLSM Calculator - Variable Length Subnet Masking Tool | SubnettingCalculator



Chapter

07 Common Networking Protocols

IP Services.

Routers and switches are equipped with various IP services that are essential for network management and operation. These services i nclude DHCP, DNS, NAT, SNMP, and NTP, which are designed to provide reliable and efficient communication over an IP network. Here' s a brief overview of each service:

- The Dynamic Host Configuration Protocol (DHCP) is a network management protocol that automatically assigns IP addresses and other communication parameters to devices on a network. It eliminates the need for manually configuring each device, streamlining the process for networks ranging from residential to large campus networks. DHCP operates using a client-server architecture, where clients request parameters from a centrally installed DHCP server.
- Domain Name System (DNS) translates human-readable domain names into IP addresses, allowing users to access websites and services using names instead of numbers. DNS servers store a directory of domain names and their corresponding IPs, facilitating easy navigation on the internet.
- Network Address Translation (NAT) translates private IP addresses used within a network to a public IP address for internet communication. This is particularly useful for devices on private networks that do not have routable IP addresses. NAT can be implemented in two ways: static and dynamic.
- Simple Network Management Protocol (SNMP) allows a network management system to collect data from devices and monitor their performance. It can also send alerts if a device goes down or behaves abnormally. SNMP is used to monitor and manage network devices such as routers, switches, servers, and printers.
- Network Time Protocol (NTP) synchronizes clocks on all devices in a network to the correct time. Accurate time is critical for logs, security, and troubleshooting. NTP clients request time from NTP servers, adjusting their clocks gradually to match the accurate time.

These protocols are essential for the proper functioning of modern networks, ensuring efficient communication and management of network devices.

Network Protocol.

PROTOCOLS (SERVICE NAMES)	PORTS NUMBERS	TRANSPORT PROTOCOLS	MEANINGS
1.File Transfer Protocol (FTP)	20 and 21	TCP	It is a protocol that carries data guarantees that data will be delivered properly.
2.Secure Shell (SSH)	22	TCP and UDP	It is a cryptographic network protocol used to secure data communication.
3.Telnet	23	TCP	It is the used for remote management protocol for managing network devices.
4.Simple Mail Transfer Protocol (SMTP)	25	TCP	It is a communication protocol which is used to transmit email messages over the internet to the destination server.
5.Domian Name System (DNS)	53	TCP and UDP	It is used in the performance of one simple task of converting IP address To domain names that everyone can easily understand.
6.Trivial File Transfer Protocol (TFTP)	69	UDP	TFTP is typically used by devices to upgrade software and firmware and that include cisco.
7.Hyper Text Transfer Protocol (HTTP)	80	TCP	It is a kind of protocol used to define how data is transmitted and formatted and also used by www as a channel for communication.
8.Dynamic Host Configuration Protocol (DHCP)	67 and 68	UDP	It is a kind of service used in the client and server model.
9.Post Office Protocol 3 (POP3)	110	TCP	It is a protocol used by e-mail client to retrieve e-mail from the servers.
10. Network News Transport Protocol (NNTP)	119	TCP	ntp is an application protocol used for transporting USENET news articles between news servers and the end user client.

Network Protocol.

PROTOCOLS (SERVICE NAMES)	PORTS NUMBERS	TRANSPORT PROTOCOLS	MEANINGS
11. Network Time Protocol (NTP)	123	UDP	It is the synchronization of time between network devices in the network.
12. NetBIOS	135 and 139	TCP and UDP	NetBIOS itself is not a protocol but is typically used in combination with IP with the NetBIOS over TCP/IP protocol.
13. Simple Network Management Protocol (SNMP)	161 and 162	TCP and UDP	It has the ability to monitor, configure and control network devices.
14. Lightweight Directory Access Protocol	389	TCP and UDP	LDAP provides a mechanism of accessing and maintaining distributed directory information.
15. Transport Layer Security (TLS)	443	TCP	It is a protocol of a secured socket layer that uses asymmetric keys to transfer data over a network.
16. Real-Time Transport Protocol. (RTP)	1023 TO 65535	UDP	It is used for delivering audio and video data over an IP network.
17. Hyper Text Transfer Protocol Secure. (HTTPS)	443	TCP	It renders authentication and encryption that provides secure communication with the use of secure socket layer.
18. Internet Message Access Protocol. (IMAP4)	143	TCP and UDP	It is an application layer protocol and an internet standards for e-mail retrieval.
19. Address Resolution Protocol (ARP)	3389	TCP	It is used to resolve the network layer address into the link address.
20. Border Gateway Protocol (BGP)	179	TCP	It is used to maintain very large routing tables and traffic processing.

Network Protocol.

PROTOCOLS (SERVICE NAMES)	PORTS NUMBERS	TRANSPORT PROTOCOLS	MEANINGS
21. Internet Relay Chat (IRC)	194	UDP	It is an application layer protocol that facilitate communication in the form of text.
22. Session Initiation Protocol. (SLP)		TCP and UDP	It is used to establish, modify, and terminate multimedia communication session such as VoIP.
23. Session Description Protocol.(SDP)		TCP	It describes the content of multimedia communication.
24. Remote Desktop Protocol. (RDP)	3389	TCP	It provides a user with a graphical interface to connect to another computer over a network connection.
25. Server Message Block (SMB)		TCP	It is an application layer protocol that helps in accessing network resources, such as shared files and printers.
26. Secure File Transfer Protocol (SFTP)	22	TCP and UDP	It uses the SSH protocol to access and transfer file over the network.
27. Internet Group Management Protocol (IGMP)	2	TCP	It is a communication protocol used by hosts and adjacent routers on IPv4 network to establish multicast group membership.
28. Route Access Protocol (RAP)	38	TCP	
29.Resource Location Protocol (RLP)	39	TCP	It is used for determining the location of higher level service from host on a network.
30. Host Name Server Protocol (HNSP)	42	TCP	

Network Protocol.

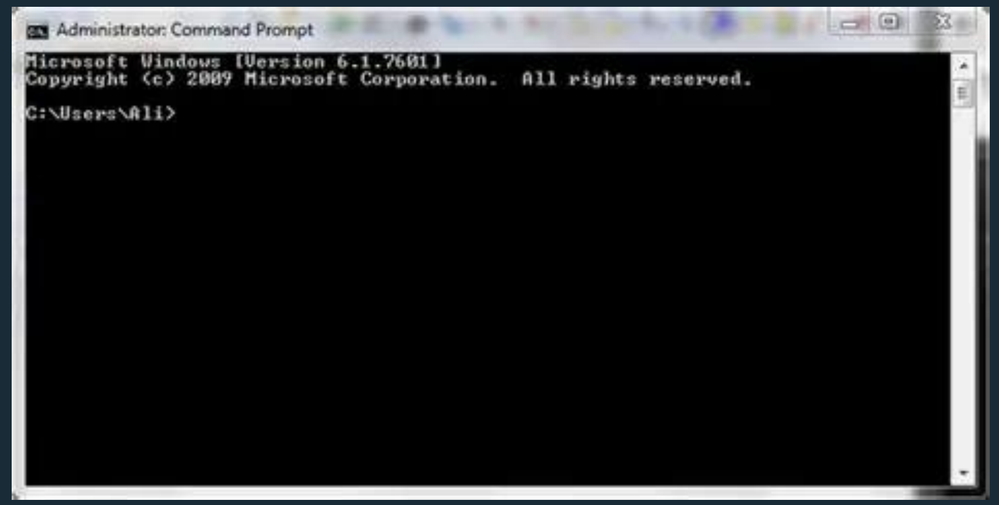
PROTOCOLS (SERVICE NAMES)	PORTS NUMBERS	TRANSPORT PROTOCOLS	MEANINGS
31.Internet Control Messages Protocol (ICMP)		PING	It is used by a ping utility to check the reachability the device in a network.
32.Remote Directory Access Protocol (RDAS)		TCP	It is used retrieves information about domain names from a central registry.
33.Lightweight Presentation Protocol (LPP)		TCP and UDP	It is describe an approach for providing stream lined support of OSI application services on top of TCP/IP –based network for some constrained environment.
34. Remote Procedure Call Protocol (RPC)		TCP and UDP	It is a protocol for requesting a service from a program location in a remote computer through a network.
35. Network Address Translation (NAT)	3022	TCP and UDP	It is the method by which IP addresses are mapped from one group to another, transparent to end users.
36. Microsoft Active Directory Protocol (MADP)	445	TCP	it is used by Microsoft server operating systems for client/server access and file and printer sharing.
37.Calender Access Protocol (CAP)	1026	TCP	It is used by Novell GroupWise for its calendar access protocol and also used by windows task scheduler.
38. Layer Two Tunneling Protocol.(L2TP)	1701	TCP	It is used to connect two private business network together over an internet connection to create a virtual network.
39.Point To Point Tunneling Protocol (PPTP)	1732	TCP	A tunneling and encryption standard is used to connect two private business network together over an internet connection to create a virtual network.
40.Remote Procedure Call (RPC)	135	TCP	It holds information regarding which ports and IP addresses the services are currently running .

Troubleshooting Application Connectivity Issues.

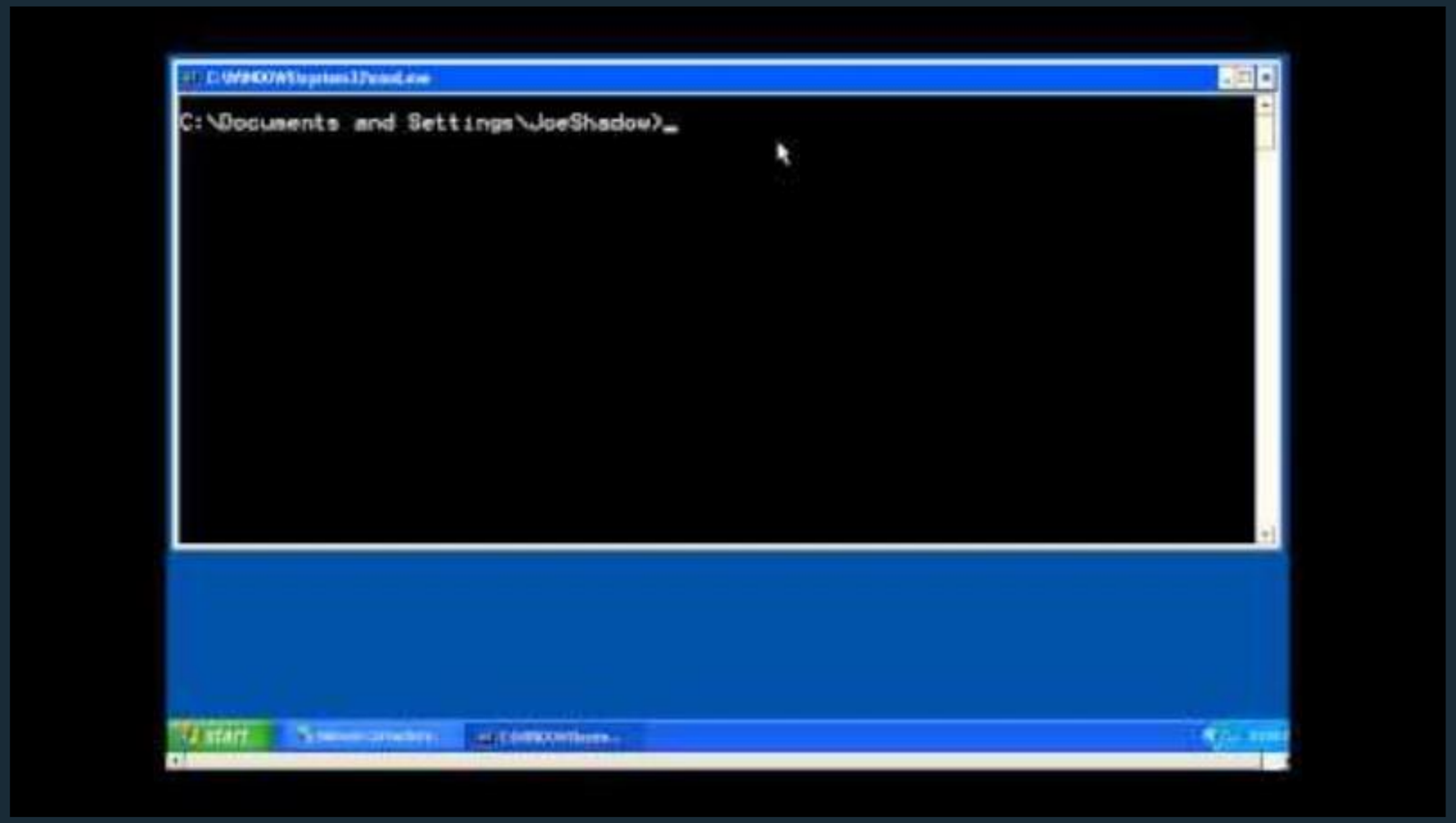
Command line tools are essential for network troubleshooting because they allow IT professionals to quickly check network connectivity, analyze traffic, and gather information about devices on the network. These tools are used in **Windows, Linux, and macOS environments**, though some commands have slight differences.

Tool	Purpose	Key Points
ping	Test connectivity	Uses ICMP, shows response time, packet loss
traceroute / tracert	Trace path to destination	Shows each hop, latency, routing issues
nslookup	DNS resolution	Forward/reverse lookup, query specific DNS
tcpdump	Packet capture & analysis	Filters traffic, real-time analysis, Linux/macOS
dig	DNS lookup	Detailed queries, record types, Linux/macOS
netstat	Connections & ports	Active connections, routing tables, protocol stats
ip / ifconfig / ipconfig	IP configuration	Shows IP, subnet, gateway; DHCP release/renew
arp	Map IP to MAC	View, add, delete ARP entries; LAN troubleshooting

Command Line Interface (CLI)



Troubleshooting Application Connectivity Issues.



Chapter

08

IPSEC VPN / SSL VPN

The difference between SSL and IPsec VPNs is that SSL VPNs secure individual web sessions, while IPsec encrypts entire network traffic. SSL VPNs are generally used for secure web application access and are easier to use because they do not require dedicated VPN client software.

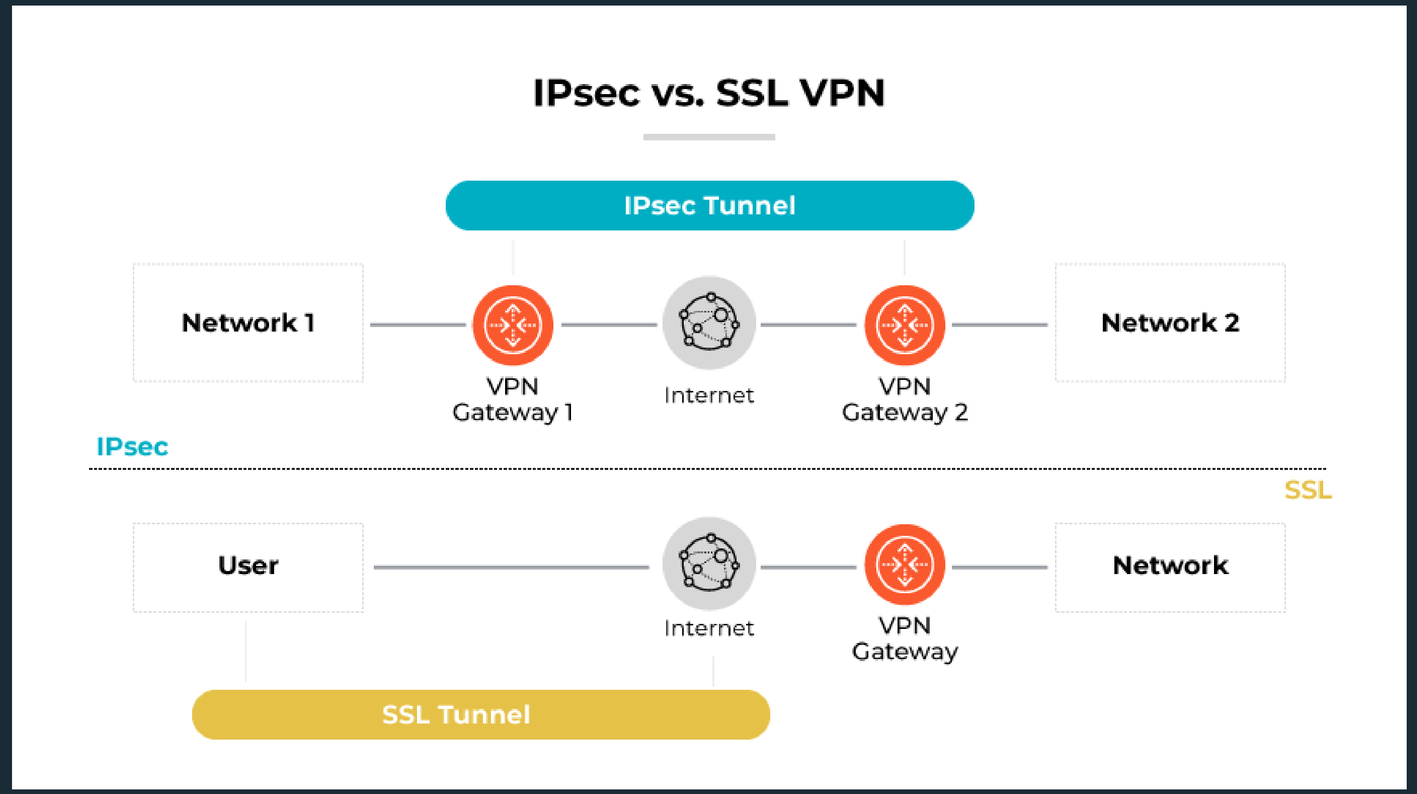
SSL VPN.

SSL, also known as Secure Sockets Layer, functions at the application layer within the OSI model. Its primary purpose is to secure and encrypt the data that is transmitted between a user’s web browser and the web server.

IPSec VPN.

IPsec, also known as Internet Protocol Security, provides security for internet communication on the network layer. It consists of a collection of protocols that encrypt and authenticate network traffic.

IPSec vs SSL VPN: A Comprehensive Comparison.



IPSec vs SSL VPN:
A Comprehensive Comparison.

IPSec vs SSL VPN: A Comprehensive Comparison.

Security/ Protection	SSL VPN	IPSec VPN
Spoofing Prevention	Server and clients provide credentials in order to confirm the identity of both systems	Recipient cannot be tricked as the identity of each end point is verified.
Modification Prevention	Every packet is signed with a hash function called HMAC which ensures that there has not been any modification to the packet.	Packet cannot be ·Captured ·Payload modified And then sent further without it being detected.
Protection	Data is encrypted by the public key which makes sure that only the recipient is able to decrypt and receive the information.	Data is encrypted and one cannot read the content of the packet payload.
Reuse Prevention	Packets cannot be re-used or any passwords or keys discovered.	Packets cannot be re-used or discover any passwords or keys.

IPSec vs SSL VPN: A Comprehensive Comparison.

Features	SSL	IPSEC
Application for	Web based applications	IP based applications
VPN Type	Client to Site only (Remote Access)	Site to Site or Client to Site (Remote Access)
Encryption	Moderate to Strong	Strong
Authentication	One way or two-way authentication	Two-way authentication using shared leys or digital certificates.
Connection Complexity	Low Requires only web browser	Medium Full configuration, end-to-end tunnel establishment
Connection Options	Any device can connect	Only specific devices
Connectivity	Connects users to specific applications and services	Connects remote hosts to entire networks
Configuration	Easy	Hard

Chapter

09 Bonus

GenAI in Infrastructure as Code.

Generative AI (GenAI) is transforming Infrastructure as Code (IaC) practices within enterprise networking environments by automating, optimizing, and simplifying infrastructure provisioning, configuration, and maintenance workflows.

GenAI in Infrastructure as Code

Practical Applications

Use Case	GenAI Role	Impact
Script Generation	Auto-generate Ansible/Terraform code	Speeds provisioning, reduces human error
Capacity Planning	Enterprise LLM predicts traffic/load	Optimized resource allocation, cost savings
Network Configuration	Auto-apply configurations	Reduces manual intervention and misconfiguration
Documentation	Auto-document IaC and APIs	Facilitates knowledge transfer, compliance
Root Cause Analysis	Analyze logs & network metrics	Faster MTTR, improved reliability

Infrastructure as Code for Generative AI Models - AWS

GenAI in Infrastructure as Code

1. GenAI-Enhanced IaC Automation

Script and Playbook Generation:

GenAI can automatically generate code artifacts for popular IaC tools such as:

- **Ansible**: Playbooks for configuration management and deployment.
- **Terraform**: Declarative scripts for cloud infrastructure provisioning.
- **Vagrant**: Scripts for local virtual machine infrastructure.

Benefits:

- **Faster code creation**: Engineers spend less time manually writing repetitive configurations.
- **Error reduction**: **AI-generated** IaC reduces manual misconfigurations and improves consistency.
- **Parameterization**: Scripts can incorporate dynamic parameters for resource types, networking IDs, and environment-specific settings, boosting flexibility.

2. IaC for GenAI Infrastructure Workloads

Optimized Resource Provisioning:

- **Large-scale AI models and LLMs require high-performance compute, low-latency storage, and high-speed networks.**
- **IaC practices, combined with GenAI, ensure repeatable and scalable deployments across cloud and on-premise systems, e.g., AWS EC2 instances, VPCs with public/private subnets, NAT gateways, and load balancers.**

End-to-End Testing and Validation:

- **Tools like pyATS and Cisco Genie can integrate virtual traffic and synthetic test workloads, allowing GenAI-driven simulation before production deployment.**

3. Operational Efficiency

Predictive Management:

- GenAI can analyze patterns in usage, resource allocation, or traffic to proactively adjust infrastructure, e.g., auto-scaling network resources, balancing loads, or optimizing storage performance.

Intelligent Root Cause Analysis:

- AI agents identify network issues and performance bottlenecks by cross-analysing logs, metrics, and historical incidents, reducing Mean-Time-To-Resolution (MTTR).

GenAI in Infrastructure as Code

4. Security and Compliance

Dynamic Configuration Management:

- GenAI-generated IaC helps enforce consistent security policies across cloud and on-premise deployments.
- Automatic generation of security groups, firewall rules, and access policies ensures compliance with corporate standards.

Vulnerability Assessment:

- Enterprise GenAI agents can automatically evaluate infrastructure for misconfigurations or emerging security risks within IaC scripts.

5. Organizational Integration (Cisco Case)

Cisco × OpenAI Example:

- Intelligent agents (e.g., OpenAI Codex integrated into Cisco CI/CD pipelines) act as **embeddable engineering assistants**:

1. Execute automated workflows (compile → test → deploy).
2. Maintain auditable and traceable infrastructure changes.
3. Facilitate cross-repository knowledge sharing and governance.

- Quantifiable improvements include ~20% reduction in build times and 10–15× improvement in defect remediation cycles.

6. Challenges and Considerations

Terraform IaC Complexity: GenAI may initially generate code incompatible with some AWS resources; human oversight is needed for provider configurations and parameter files.

Human-in-the-loop Governance: Despite automation, domain expertise is critical to validate outputs, especially in hybrid or legacy infrastructure environments.

Scalability and Reliability: High-demand GenAI workloads require careful orchestration to prevent API sprawl, resource contention, or security gaps.

endava

DevOps & Cloud Foundations

Thank you.